

Reference Shock-Avoiding Figure-3 Campaign Claim-Limited Handoff

AthenaK Z4c investigation

28 August 2026

Final verdict

REFERENCE_GAUGE_IMPLEMENTATION_INCOMPLETE

The exact literature gauge was identified and mapped to AthenaK. The existing lapse right-hand side already implements

$$\partial_t \alpha = \beta^i \partial_i \alpha - (\alpha^2 + \kappa)(\hat{K} + 2\Theta), \quad \kappa = 1, \quad \beta^i = 0.$$

The base source nevertheless rejected nonpositive lapse in the timestep and stage-admissibility checks. This is incompatible with the reference shock-avoiding slicing, which permits lapse zero crossing.

Narrow source correction

The patch allows any finite lapse only when `lapse_shock_avoiding=true`, retains the positive-lapse requirement for every other gauge, and uses $|\alpha|$ only in the physical light-cone CFL speed. The gauge speed remains

$$v_{\text{gauge}} = \sqrt{(\alpha^2 + \kappa)\gamma^{nn}}.$$

No floor, clipping, regularization, gauge retuning, AMR-transfer change, or boundary change was introduced.

Focused verification

Five host unit/static checks passed: state admissibility and its static audit, timestep contract and its static audit, and the shock-avoiding-gauge static test. These checks establish only the narrow source contract.

Production and convergence status

No Perlmutter CUDA run and no N256 evolution were executed. The prepared N256 input is configuration evidence only. No fresh AMR authority exists, the N256 Figure-3 gate was neither passed nor failed, and N128/N512 replay was not attempted. There is no stability, Figure-3 reproduction, numerical-failure, or convergence claim.

Remaining limitations

Perlmutter CUDA qualification, the fresh N256 evolution to the published peak/minimum/rebound region, direct unshifted curve comparison, terminal failure localization if needed, and conditional three-resolution replay remain outstanding. The accompanying JSON manifest carries exact hashes and the same claim boundary.